STADIUM-DT Synthetic Game-Day Dataset: A Data Descriptor

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Abstract

This dataset records one simulated American-football game day on a stadium concourse, generated by the STADIUM-DT crowd digital twin testbed, together with machine-readable copies of the results reported in the STADIUM-DT paper. Part 1 covers 18:30 to 22:59 at one-minute resolution: 2,700 zone records for ten concourse zones, 1,080 gate records, 1,080 amenity records and 191 events. Each zone record holds the simulator's ground-truth occupancy next to the twin's fused estimate, its confidence, and the weight each input (camera, WiFi, model) carried. Part 2 holds 11 CSV tables with the paper's results for experiments E1 to E7 and their configurations. Everything is synthetic; nothing was measured at Beaver Stadium or any other venue. The day has three crowd peaks: pre-game arrivals (about 1,570 people at 19:15), the halftime amenity surge (about 1,550 at 20:59) and post-game egress (about 1,180 at 22:37). Estimates run 2.2% low on average, and the twin underestimates most where density is highest: in the 13 zone-minutes when true density reached 3.5 people/m², it reported HIGH in only 6, at an unchanged confidence of about 0.90.

Background and summary

Crowd digital twins keep a model of where people are in sync with sensors, so operators can see current conditions, forecast congestion and test interventions before acting. Real crowd data with ground truth is scarce: venues rarely release it, cameras and WiFi are privacy-sensitive, and nobody counts every person in every zone. Synthetic data from a known model fills that gap for method development, as long as its origin and limits are explicit.

STADIUM-DT is a crowd digital twin design for one concourse of Beaver Stadium. Only aggregate counts, flows, speeds and queue lengths reach the twin, and one zone-level crowd-flow model serves as fusion prior, forecaster, scenario engine and router. Its browser-based testbed simulates the venue, emulates cameras, WiFi access points and turnstiles with realistic error models, and runs the twin on top.

This descriptor documents a dataset produced by running that testbed through one game day, plus the results the STADIUM-DT paper reports from its experiments. The value of Part 1 is that every zone-minute pairs the true occupancy with what the twin estimated

from imperfect sensors, so estimation error, bias and confidence can be studied directly. The sections below explain how the data was generated, what each file holds, what happens during the day, how far the data can be trusted and how to use it.

Methods

The data comes from the released STADIUM-DT testbed, a single HTML page whose JavaScript model was run unchanged; only the 3D rendering library was replaced by a stub, which does not touch the crowd model.

Simulated venue. The pilot is a 110 m stretch of the west concourse split into ten zones, CONC-W-010 to CONC-W-019, with walkable areas of 62 to 77 m². It connects to Gates A, B and C (8, 10 and 8 screening lanes) and an auxiliary Gate D (6 lanes, closed by default), three portals to the lower bowl, two stairs, an elevator, a ramp, two concessions, two restrooms and a first-aid station. It serves 9,600 seats at 96.5% simulated attendance. The layout and labels are a stylised concept, not survey data.

Crowd model (ground truth). People move as streams between those attachments. Arrivals peak 72 minutes after the 18:00 gate opening at 201 people/min; gates screen 6.5 people per lane per minute and queue the rest. Spectators choose amenities and exits with a logit rule that weighs walking time and expected wait. Zone occupancy follows Little's law with a density-dependent walking speed, capacity-limited backlogs and queue spillover. The run uses the model's nominal parameters, so this ground truth is the testbed's default game day.

Synthetic sensors and fusion (estimate). Every second, emulated cameras report each zone's count with a density-dependent undercount (2% plus 3.5% per person/m² above 2, so about 7% at 3.5 people/m²) and about 5% correlated noise. WiFi access points report device counts that lag up to 30 s and drift with the device-to-person ratio. Turnstiles report entries exactly. The twin fuses camera, WiFi and a model prior per zone by inverse-variance weighting, which gives the estimate, its confidence and the three weights stored in the data. Anomaly rules run every 2 s on the estimate.

Simulation protocol. The twin advanced in 0.5 s steps and every zone, gate and amenity was recorded once per simulated minute. The day was run in two segments:

1. **A, 18:30 to 22:33.** Initialised at 18:30 (the testbed integrates from gate opening at 18:00) and run continuously through the first half, halftime and second half.
2. **B, 22:34 to 22:59.** Re-initialised at the final whistle with the seated catchment released, as the testbed does when its POST-GAME phase is selected. A continuous run never starts post-game egress by itself, so without this step the post-game hour would show an empty concourse. The re-initialisation makes the concourse jump from about 216 to 20 people at 22:34.

Paper results (Part 2). The result and configuration tables of the STADIUM-DT manuscript were extracted programmatically into tidy CSV files. Values were copied, not recomputed; cells that combined several numbers were split into one value per row.

Data records

The package (stadium_dt_dataset_v2.zip) holds 22 files in four parts: a README, `synthetic_game_day/` (4 CSV files), `paper_results/` (11 CSV files) and `code/` (the generation script and rendering stub). All files are comma-separated with a header row; times are simulated clock times (HH:MM:SS) on one game day.

Part 1: synthetic game day

File	Rows	One row is	Key columns
zones.csv	2,700	one zone at one minute (10 × 270)	true vs estimated people, density, speed, flows, confidence, input weights
gates.csv	1,080	one gate at one minute (4 × 270)	open, throughput, queue
amenities.csv	1,080	one concession or restroom at one minute (4 × 270)	queue, queue capacity
events.csv	191	one alert, forecast or clearing, to the second	category, text

All files carry `time`, `phase` (PRE-GAME, 1ST HALF, HALFTIME, 2ND HALF, POST-GAME) and `segment` (A or B, see Methods).

Column (zones.csv)	Unit	Meaning
zone_id	—	CONC-W-010 to CONC-W-019, south to north
area_m2	m²	walkable area, 62 to 77
true_people	people	simulator ground truth; not observable in a real venue
estimated_people	people	twin's fused estimate from emulated sensors

Column (zones.csv)	Unit	Meaning
density_ppm2	people/m²	estimated density (unrounded estimate ÷ area)
density_level	—	LOW < 1.5 ⩽ MODERATE < 2.5 ⩽ ELEVATED < 3.5 ⩽ HIGH
speed_mps	m/s	estimated mean walking speed
flow_direction	—	dominant direction (NORTHBOUND, EASTBOUND, …, STATIONARY)
inflow_per_min, outflow_per_min	people/min	flows across the zone boundaries
amenity_queue	people	queue at an amenity inside the zone
confidence	0.25–0.97	confidence score shown to operators
weight_cctv, weight_wifi, weight_model	share	each input's share of the fused estimate; they sum to 1

In gates.csv, `throughput_per_min` is people screened per minute and `queue_people` the estimated queue; Gate D stays closed all day. In amenities.csv, `queue_capacity` is the space before a queue spills into the concourse (40 at concessions, 30 at restrooms). In events.csv, `category` is `alert` (190 rows) or `predict` (1 bottleneck forecast), and the text names the alert and zone; 96 rows open an alert or forecast and 95 mark an alert as "Cleared".

Part 2: paper results

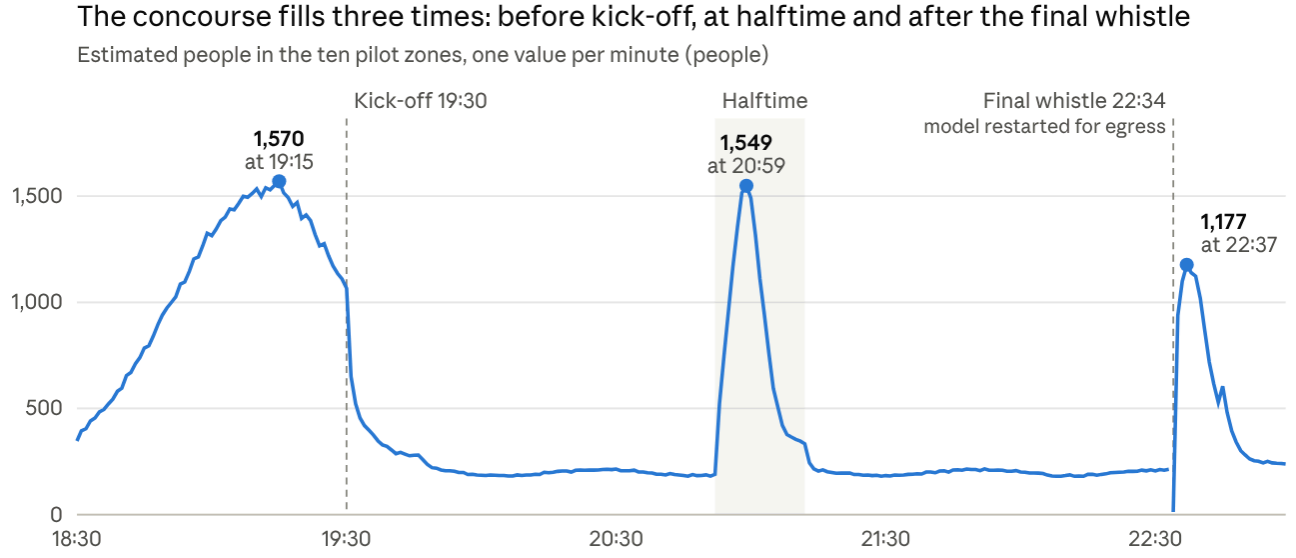
File	Content	Rows
E1_fusion_accuracy.csv	error, 95% interval coverage and lag by prior, sources and window	36
E2_forecast_accuracy.csv	forecast error, skill and threshold-event F1 by method	6
E3_detection.csv	detections of 19 episodes, latency, false alarms by alarm rule	12
E4a_single_requests.csv	crowd-aware vs shortest routes by destination	6
E4b_burst_requests_K120.csv	time to be served and concentration by policy and variant	18

File	Content	Rows
E5_E6_scenarios.csv	clearance and peak-density effects of nine interventions under three conditions	27
E7_compute_cost.csv	run time of twin functions	10
scenario_definitions.csv	the nine what-if scenarios	9
ground_truth_perturbations.csv	random variations applied per experiment replicate	5
boundary_messages.csv	messages crossing the aggregate-only boundary	5
validation_targets.csv	proposal targets against the evidence	5

These are summary statistics: means over 10 to 40 replicates or pooled counts, exactly as reported in the paper.

What happens in the data

The day has three crowd peaks of 1,100 to 1,600 people, separated by long quiet stretches when spectators are in their seats.



zones.csv · estimated_people summed over 10 zones · 18:30–22:59, 270 minutes

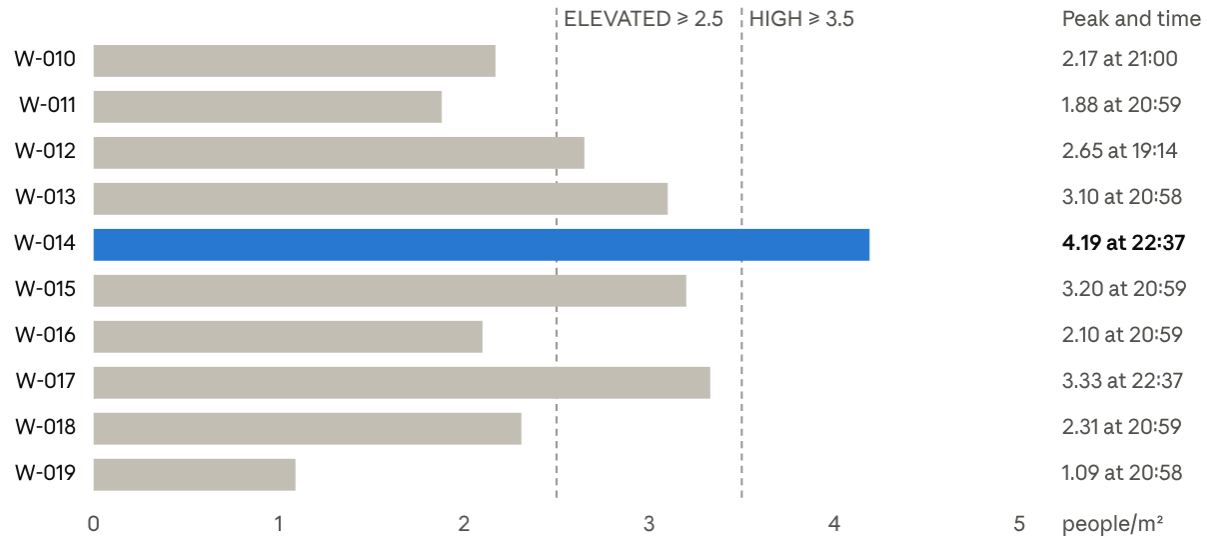
Arrivals build from 18:30 and peak at 19:15, then the concourse empties within five minutes of kick-off. During play it holds about 180 to 215 people. Halftime pulls about 1,550 people out of the seats within seven minutes. After the final whistle, egress peaks at about 1,180 people three minutes later and falls below 300 people by 22:50.

Where the crowd concentrates

The middle of the concourse, W-013 to W-015 by Gate B and the 104/105 corridor, carries the load; the two ends stay below 2.2 people/m² all day.

Only W-014 reaches HIGH density, peaking at 4.19 people/m² after the final whistle

Highest estimated density of each zone during the day, zones in order along the concourse

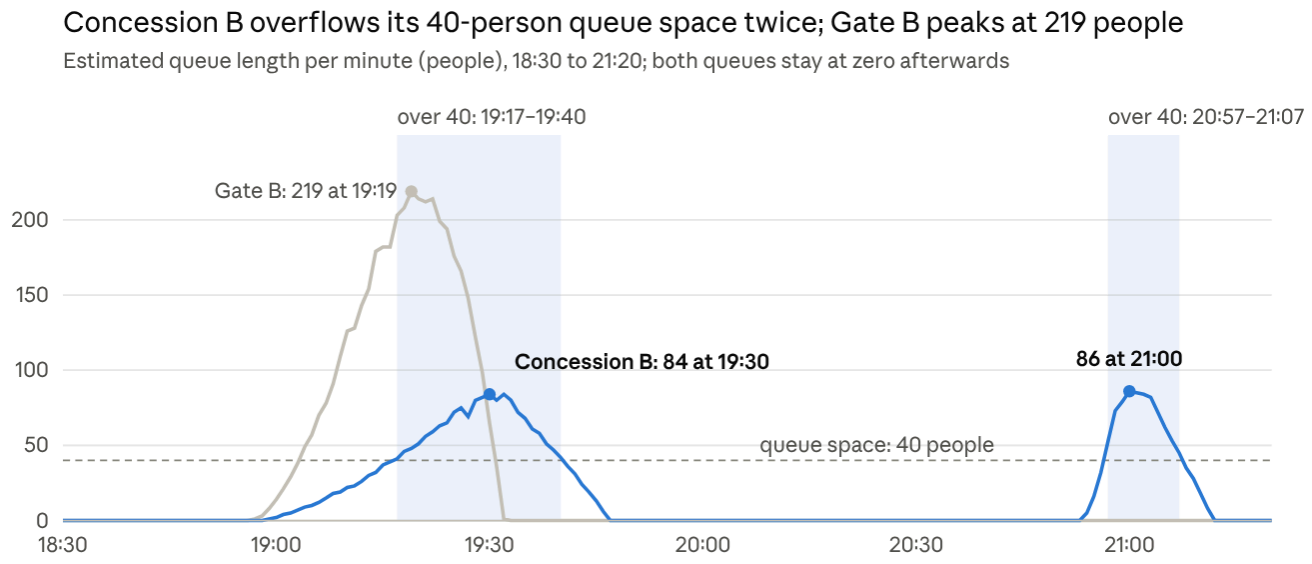


zones.csv · maximum of density_ppm2 per zone · 18:30–22:59

W-014 is the narrowest zone (62 m²) and sits between Gate B and the main portal, so arrivals and egress both pass through it. It is ELEVATED for 31 minutes and HIGH for 6, all six between 22:35 and 22:40. Across all zones the day has 2,282 LOW, 338 MODERATE, 74 ELEVATED and 6 HIGH zone-minutes; ELEVATED minutes fall mostly before kick-off (59) and at halftime (10).

Queues

Gates queue before kick-off and concessions overflow twice, while restrooms never exceed their space.



amenities.csv (CONC_B) and gates.csv (B) · queue_people · 18:30–21:20

All three open gates queue from about 18:57 to 19:32, peaking together at 19:19 (Gate A 136, B 219, C 112 people); Gate D stays closed and the gates screen about 8,800 entrants in total. Both concessions overflow their 40-person space before the game and again at halftime, peaking at 85 (A) and 86 (B) people at 21:00; the overflow spills into the adjacent concourse zones. Restroom queues peak at 8 (A) and 23 (B) people against space for 30.

Alerts

The alert log mirrors the crowd peaks: 67 of the 96 openings fall in the 19:00 hour and 16 in the post-game window.

Alert type	Openings
Congestion developing	71
Counter-flow detected	8
Capacity constraint	6
Queue spillover (concessions)	4
High density (all in W-014)	3
Stair bottleneck	2
Movement slowdown	1
Bottleneck forecast	1

Most openings name W-014 (29), W-015 (22) and W-013 (19).

Technical validation

The files are complete and internally consistent, and the twin's estimate tracks the truth closely (correlation 0.998) but runs low, most of all where density is highest.

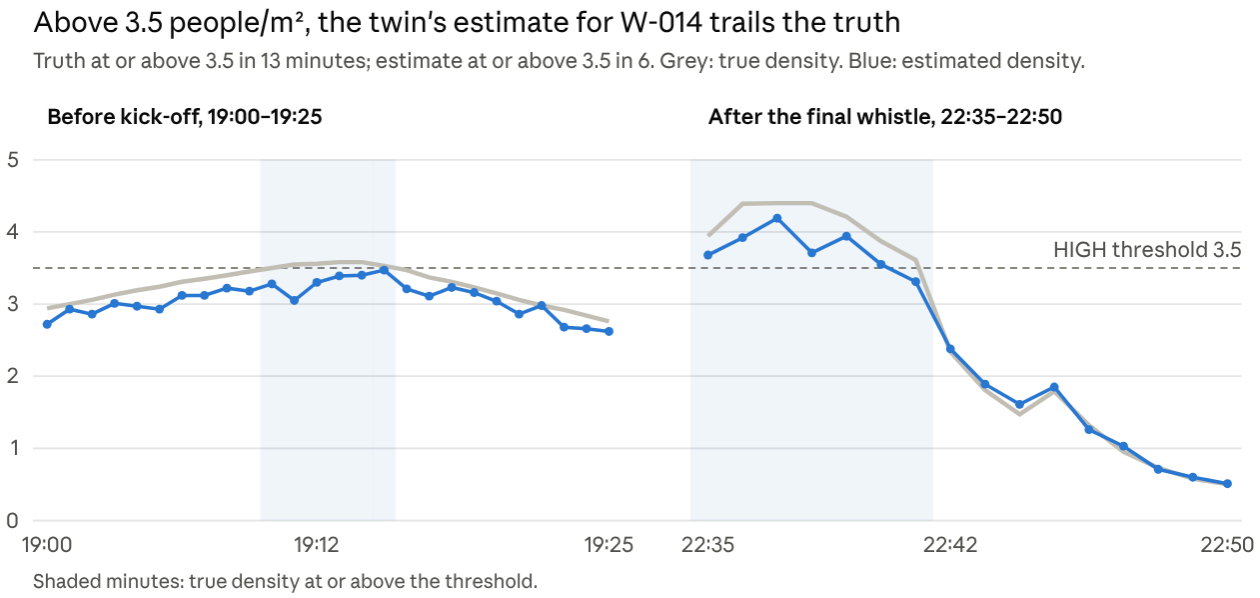
Integrity checks on Part 1, all passed:

Check	Result
Missing values	none in any file
Time grid	270 minutes, 18:30 to 22:59, exactly 10 zones, 4 gates and 4 amenities per minute
Density = estimate ÷ area	agrees within 0.012 people/m² (rounding of the stored estimate)
Density level matches thresholds	99.9% of rows; the rest sit on a threshold and differ only by rounding
Input weights	sum to 1.00 ± 0.01 in every row
Part 2 values	extracted programmatically and compared with the manuscript tables

Estimate against truth. Over the 1,700 zone-minutes with at least 20 people, the estimate misses the true occupancy by 3.8% on average and is 2.2% low. Pre-game is the hardest phase (4.4% error, 3.6% low); the other phases stay at 3.2% to 4.0%.

True density (people/m²)	Zone-minutes	Mean error	Mean bias	Mean confidence
below 1.5	1,265	3.6%	−1.8%	0.84
1.5 to 2.5	316	4.2%	−2.9%	0.90
2.5 to 3.5	107	5.1%	−4.7%	0.90
3.5 and above	12	7.9%	−7.9%	0.91

The bias grows with density while confidence stays flat at about 0.90, so the confidence score does not warn when the estimate is least reliable. This is the camera-undercount effect the STADIUM-DT paper reports in its experiment E1, here visible in a single run.



zones.csv · CONC-W-014 · true density = true_people ÷ area_m2; estimated = density_ppm2

Before kick-off, true density in W-014 stays at or above 3.5 for six minutes (19:10–19:15) while the estimate peaks at 3.47, so no one-minute snapshot shows HIGH; the alert rules, which run every 2 s, still fired two HIGH DENSITY alerts at 19:12:49 and 19:15:11. After the final whistle the truth reaches 4.40 and the estimate 4.19, and the estimate shows HIGH in 6 of the 7 minutes the truth does.

Usage notes

The dataset suits work that needs a known truth next to an imperfect estimate:

- testing anomaly or congestion-detection rules against `true_people`, and measuring missed or late alarms
- studying sensor-fusion bias and confidence calibration by density band
- prototyping short-horizon forecasts of zone density or queue length
- teaching crowd-flow concepts: arrival peaks, queue spillover, halftime surges, egress
- building and demonstrating dashboards with realistic but harmless data

To load it in Python: `pd.read_csv("synthetic_game_day/zones.csv")`, then pivot on `zone_id` for a time × zone matrix. To join gates or amenities, use `time`.

Four pitfalls:

1. Use `estimated_people` for anything an operator could see; `true_people` exists only in simulation.
2. Split analyses at 22:34: segment B starts from a re-initialised state, so the jump there is an artefact, not a crowd event.

3. One-minute snapshots miss short events: two HIGH DENSITY alerts fired between snapshots before the game, while no snapshot shows HIGH then.
4. Part 2 values are means over replicates, not per-run data; do not treat them as observations.

Limitations

- **Synthetic only.** No value describes a real venue or real spectators. Layout, capacities, arrival profile and behaviour are design assumptions of the testbed, not Beaver Stadium data.
- **One day, one run.** Part 1 is a single nominal game day. It shows no day-to-day variation and none of the perturbed demand, timing or capacity used in the paper's experiments, so its error figures are not the paper's E1 results.
- **Same model for truth and twin.** The truth comes from the model the twin also uses, and the twin's prior uses the true increments; real estimation errors would likely be larger.
- **Assumed sensor errors.** The camera undercount, WiFi drift and noise levels are plausible assumptions, not measurements.
- **Coarse sampling.** One snapshot per minute; raw per-second sensor readings are not stored.
- **Stitched post-game.** Segment B restarts the model at the final whistle (Methods).
- **No interventions.** Gate D stays closed and no scenario runs; intervention effects appear only in the Part 2 summaries.
- **Summaries in Part 2.** Per-replicate experiment outputs are not included. Three reported values (TWIN_IR skill 0.73, pilot-only feedback gains of 2.5% and 2.7%, real-time factor 2,768) are flagged in the paper's supplement for author confirmation.

Data and code availability

The package is stadium_dt_dataset_v2.zip, delivered with this document; it has no public identifier yet. Part 1 was produced by `code/generate_synthetic_game_day.js`, which loads the released STADIUM-DT testbed page (version 1.0) in headless Chromium through Playwright, stubs the three.js renderer with `code/three_stub.js`, and writes the four CSV files. The testbed seeds its random numbers, but the page draws some while loading, so a rerun in another environment may differ slightly in the noise. Part 2 was extracted from the STADIUM-DT manuscript (revision R2), whose IEEE conference version reports the same values. Version 1 of the package, which lacked post-game egress and stepped the model in 1 s steps, is superseded.

Open question: where should the package be deposited (for example Zenodo or a university repository) so it can be cited with a DOI?